

Q1: 24 Feb (Shift 2) - Numerical

Two solids A and B of mass 1 kg and 2 kg respectively are moving with equal linear momentum. The ratio of their kinetic energies $(K.E.)_A : (K.E.)_B$ will be $\frac{A}{1}$. So the value of A will be

Q2: 25 Feb (Shift 1) - Numerical

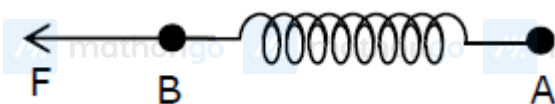
The potential energy (U) of a diatomic molecule is a function dependent on r (interatomic distance) as $U = \frac{\alpha}{r^{10}} - \frac{\beta}{r^5} - 3$ Where, a and b are positive constants. The equilibrium distance between two atoms will be $\left(\frac{2\alpha}{\beta}\right)^{\frac{a}{b}}$. Where a =

Q3: 25 Feb (Shift 2) - Numerical

Two particles having masses 4 g and 16 g respectively are moving with equal kinetic energies. The ratio of the magnitudes of their linear momentum is n : 2. The value of n will be

Q4: 26 Feb (Shift 2) - Single Correct

Two masses A and B, each of mass M are fixed together by a massless springs. A force acts on the mass B as shown in figure. If the mass A starts moving away from mass B with acceleration 'a', then the acceleration of mass B will be:



- (1) $\frac{F+Ma}{M}$
- (2) $\frac{F-Ma}{M}$
- (3) $\frac{Ma-F}{M}$
- (4) $\frac{MF}{F+Ma}$

Answer Key**Q1 (2)****Q2 (1)****Q3 (1)****Q4 (2)**

Q1 (2)

Given that, $\frac{M_1}{M_2} = \frac{1}{2}$

Also, $p_1 = p_2 = p$

$\Rightarrow M_1 V_1 = M_2 V_2 = p$

Also, we know that

$K = \frac{p^2}{2M} \Rightarrow K_1 = \frac{p^2}{2M_1} \& \Rightarrow K_2 = \frac{p^2}{2M_2}$

$\Rightarrow \frac{K_1}{K_2} = \frac{p^2}{2M_1} \times \frac{2M_2}{p^2} \Rightarrow \frac{K_1}{K_2} = \frac{M_2}{M_1} = \frac{2}{1}$

$\Rightarrow \frac{A}{1} = \frac{2}{1} \Rightarrow \therefore A = 2$

Q2 (1)

$F = -\frac{dU}{dr}$

$F = -\left[-\frac{10\alpha}{r^{11}} + \frac{5\beta}{r^6}\right]$

for equilibrium, $F = 0$

$\frac{10\alpha}{r^{11}} = \frac{5\beta}{r^6}$

$\frac{2\alpha}{\beta} = r^5$

$r = \left(\frac{2\alpha}{\beta}\right)^{1/5}$

$a = 1$

Q3 (1)

$\therefore$ relation b/w kinetic energy & momentum is

$P = \sqrt{2mKE} \quad (\because KE = \text{same})$

$\frac{p_1}{p_2} = \sqrt{\frac{m_1}{m_2}}$

$\frac{n}{2} = \sqrt{\frac{4}{16}}$

$n = 1$

Q4 (2)



$$F - F_s = Ma'$$

$$a' = \frac{F}{M} - 9$$

$$\frac{F - ma}{M}$$